NASA BIOGRAPHICAL SKETCH FORM

Identifying Information

Name: Goodrich, Colton, Lynn

Persistent Identifier (PID): ORCID 0009-0006-5237-2836 (https://orcid.org/0009-0006-5237-2836)

Position Title: Doctoral Researcher, Geosensing Systems Engineering & Sciences

Organization and Location: University of Houston, Houston, TX, USA

Professional Preparation

University of Houston — Houston, TX, USA

Degree: Ph.D., Geosensing Systems Engineering & Sciences. Start: 08/2025. Received: Expected 08/2029. Field: Geosensing / Remote Sensing

Weber State University — Ogden, UT, USA

Degree: Certificate, Geospatial Analysis. Start: 2018. Received: 2019. Field: Geospatial Analysis / GIS

Brigham Young University — Provo, UT, USA

Degree: M.S. Start: 09/2011. Received: 12/2013. Field: Geology

Brigham Young University — Provo, UT, USA

Degree: B.S. Start: 2006. Received: 2011. Field: Geology

Appointments and Positions

2025–Present · Doctoral Researcher

University of Houston, Geosensing Systems Engineering & Sciences · Houston, TX, USA

2020–2025 · Environmental Scientist III (promoted from Environmental Scientist II)

Utah Department of Natural Resources, Division of Oil, Gas & Mining · Syracuse, UT, USA

2013–2016 · Earth Scientist

Chevron Energy Company · Houston, TX, USA

2012–2012 · Reservoir Modeling Intern

Chevron Energy Company · Houston, TX, USA

2011–2013 · Graduate Teaching Assistant

Brigham Young University, Department of Geology · Provo, UT, USA

Products

1. Goodrich, C. (2025). WellSight: An autonomous lidar analysis pipeline for detecting orphaned and abandoned oil & gas wells [software]. Multi-task deep-learning system (U-Net, ~8M parameters) segmenting well pits, access roads, and well pads from lidar-derived rasters at 0.5–1 m resolution; integrates PDAL, WhiteboxTools, GDAL/Rasterio, and PyTorch. https://github.com/clgoodrich/lidar_project

2. Goodrich, C. (2025). SAOCOM L-band InSAR digital elevation model generation and validation framework [software/dataset]. P-SBAS time-series interferometry of SAOCOM-1A/1B data producing high-coherence, vegetation-penetrating DEMs; accuracy assessment (NMAD, RMSE, CORINE land-cover-stratified) against TINITALY lidar and Copernicus GLO-30, in support of the NASA Surface Topography and Vegetation (STV) Decadal Survey. https://github.com/clgoodrich/saocom_project

3. Goodrich, C. (2025). Coastal land-subsidence mapping from Small Baseline Subset (SBAS) InSAR time series [software]. Deformation-rate estimation on HyP3 SAR processing outputs with an accuracy-validation workflow. https://github.com/clgoodrich/Final_project_coastal

4. Goodrich, C. (2024). EToolsV3 — Regulatory GIS web application for directional-survey trajectory computation and Well Completion Report generation [software]. Minimum-curvature trajectory recomputation, PLSS location, boundary-footage calculation, LLM-assisted PDF extraction, interactive 2D/3D plats (FastAPI, NiceGUI, Shapely/GeoPandas). https://github.com/clgoodrich/EToolsV3

5. Goodrich, C. (2024). ArcGIS geoprocessing toolbox for social-media geospatial analysis [software]. Parameterized ArcPy tools (kernel-density hotspots, H3 hexagon aggregation, time-enabled mosaics). https://github.com/clgoodrich/tweet_project_v2

6. Goodrich, C. (2024). 1 m DEM/hillshade mosaic over Oil Creek State Park, Pennsylvania [dataset]. Contiguous 22-tile (~7.5 × 9 km) mosaic assembled via a greedy contiguous-patch downloader over the USGS TNM Access API. https://github.com/clgoodrich/lidar_project

7. Goodrich, C. (2024). OGM Board regulatory mapping and visualization application [software]. Decision-support application presenting well data, analyses, and digital maps for monthly hearings of the Utah Board of Oil, Gas & Mining. https://github.com/clgoodrich/OGMBoardProject

Certification

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I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Signature: Colton Goodrich Date: 07/14/2026

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